



IILM UNIVERSITY, GREATER NOIDA

School of Computer Science and Engineering

Internship Presentation

Data Analytics with Python & Power BI

STUDENT NAME

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ENROLLMENT / ROLL NUMBER

2410030691

PROGRAM & BATCH

B.Tech CSE | Batch 2024-28

ORGANIZATION

EduSkills Academy

DURATION

8 Weeks

CERTIFICATE ID

2026-9ACA033384

EduSkills Academy



Industry-Academia Collaboration

- A skilling and industry-academia collaboration initiative preparing a digitally skilled, Industry 4.0-ready workforce among Indian students.
- Partners with academic institutions to deliver structured, certification-based internship programs.
- Covers emerging technology domains including data analytics, cloud computing, artificial intelligence, and cybersecurity.
- Aims to bridge the gap between classroom learning and real industry requirements.

Introduction & Problem Statement

INTRODUCTION

An 8-week internship in Data Analytics with Python & Power BI, conducted by EduSkills Academy — covering the complete analytics workflow from data collection and cleaning, through exploratory analysis in Python, to interactive dashboarding and reporting in Microsoft Power BI.

Certificate ID: 2026-9ACA033384 | Issued 28 Jul 2026

PROBLEM STATEMENT

Organizations generate large volumes of transactional and behavioural data but often lack the in-house capability to clean, analyze, and visualize it for timely decisions. Raw data across spreadsheets, databases, and applications stays underutilized without systematic processing.

The internship addressed this by training interns to build an end-to-end analytics pipeline — Python for preparation, SQL for extraction, Power BI dashboards for decision-making.

Project Objectives

1

Set up and use a Python-based environment for data analysis, including core programming concepts

2

Clean, preprocess, and structure raw datasets using the Pandas library

3

Learn and apply SQL for data extraction, querying, and integration with Python

4

Design interactive dashboards and reports in Power BI, including data modeling and DAX

5

Integrate Python and Power BI for advanced visualization and automated reporting

6

Apply these skills on real-world style projects, culminating in a capstone dashboard

Scope & Technologies Used

SCOPE OF THE PROJECT

Covered the complete lifecycle of a data analytics project — environment setup, data cleaning, SQL querying, exploratory & visual analysis in Python, and BI dashboarding in Power BI. Concluded with two real-world style projects and a capstone E-commerce Sales & Customer Insights Dashboard.

TECHNOLOGIES & TOOLS

Python

Programming language, incl. the Pandas library for data manipulation

SQL

Fundamentals to advanced techniques and data extraction

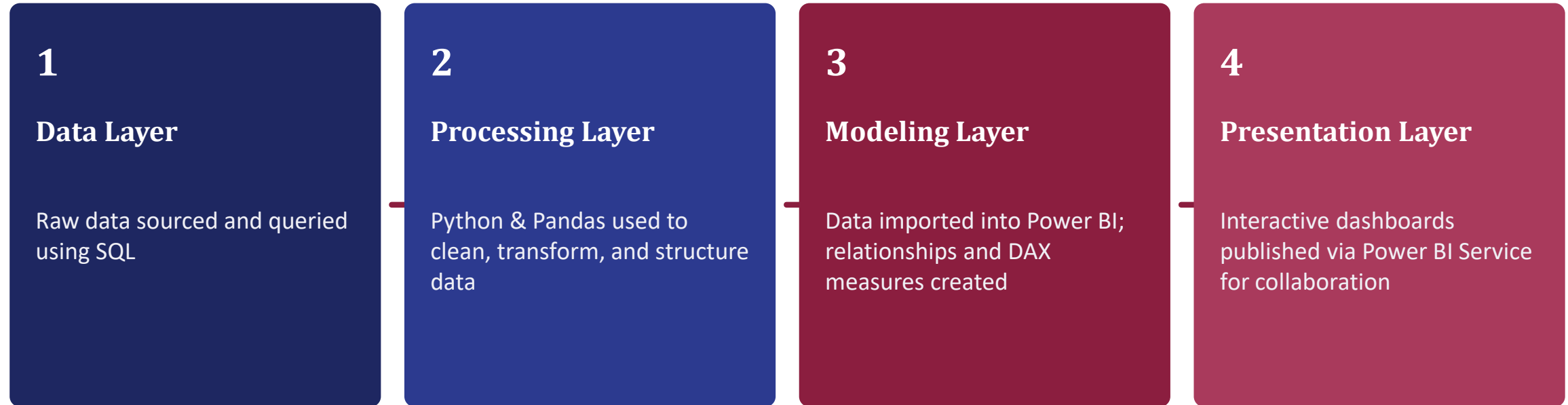
Power BI

Data modeling, DAX, visualizations, and Power BI Service

Integration

Python–SQL and Python–Power BI integration for automated workflows

System Architecture



A layered flow from raw data to business-ready insight, mirroring the internship's Python → SQL → Power BI pipeline.

Methodology — Week-wise Curriculum

W1

Environment Setup & Core Python

W2

Data Cleaning (Pandas) & SQL Fundamentals

W3

Advanced SQL & Power BI Basics

W4

Python–SQL Integration & DAX Basics

W5

Advanced Power BI & Python Visualization

W6

Python–Power BI Integration & Power BI Service

W7

Real-World Projects: Sales & Customer Behaviour

W8

Capstone: E-commerce Sales & Insights Dashboard

Real-World & Capstone Projects

1

Sales Performance Analysis

Real-world style project analyzing sales trends and performance metrics using Python and Power BI.

2

Customer Behaviour Analysis

Explored customer behaviour patterns to support data-driven business decisions.

3

Capstone: E-commerce Sales & Customer Insights Dashboard

End-to-end dashboard consolidating all eight weeks of skills — from raw data to interactive Power BI insights.

Expected Outcomes

- ✓ Ability to independently clean, preprocess, and analyze raw datasets using Python and Pandas
- ✓ Proficiency in writing SQL queries for data extraction and integration with analytical tools
- ✓ Capability to build data models and interactive dashboards using Power BI and DAX
- ✓ Practical experience analyzing sales performance and customer behaviour data
- ✓ A completed capstone project demonstrating end-to-end analytics competency

Internship Completion Certificate



Certificate ID: 2026-9ACA033384 | Issued on 28 July 2026 | Signed by Mitu Swain, General Manager – Learning & Development, EduSkills